

Derivation of the weak-form GGA exchange–correlation potential matrix used by `assemble_exc!`

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Abstract

This note derives, from the radial finite-element density parametrization used throughout `AtomicKohnSham`, the closed-form weak-form expression for the GGA exchange–correlation potential matrix $V_{xc\sigma} = \partial E_{xc}/\partial D^\sigma$ implemented in `assemble_exc!` (see `dev/gga/PLAN.md` for the original, more compressed derivation). Every factor of 2 and 4 is re-derived explicitly, since this is exactly the class of bookkeeping error that is easy to get wrong and hard to notice (a discrepancy here degrades accuracy smoothly rather than crashing).

1 Setup and notation

For a fixed angular-momentum channel and spin $\sigma \in \{\uparrow, \downarrow\}$, the radial density matrix D^σ (symmetric, $D_{ij}^\sigma = D_{ji}^\sigma$) parametrizes the reduced radial density through the finite-element basis $(\varphi_i)_i$ of the functions $u(r) = rR(r)$:

$$q^\sigma(r) = \sum_{ij} D_{ij}^\sigma \varphi_i(r) \varphi_j(r), \quad \rho^\sigma(r) = \frac{q^\sigma(r)}{4\pi r^2}. \quad (1)$$

The spin-polarized GGA exchange–correlation energy is

$$E_{xc} = \int_0^\infty e_{xc}(\rho_\uparrow, \rho_\downarrow, \sigma_{\uparrow\uparrow}, \sigma_{\uparrow\downarrow}, \sigma_{\downarrow\downarrow}) 4\pi r^2 dr, \quad \sigma_{\alpha\beta}(r) := \rho'_\alpha(r) \rho'_\beta(r), \quad (2)$$

where e_{xc} is the (Libxc-convention) exchange–correlation energy density, $v_{\rho\alpha} := \partial e_{xc}/\partial \rho_\alpha$ and $v_{\sigma\alpha\beta} := \partial e_{xc}/\partial \sigma_{\alpha\beta}$.

The Kohn–Sham weak form requires the Hamiltonian matrix element contributed by E_{xc} , obtained by differentiating E_{xc} with respect to each entry of $D^\uparrow$ (the $\downarrow$ channel follows by symmetry):

$$V_{xc\uparrow} := \frac{\partial E_{xc}}{\partial D^\uparrow}. \quad (3)$$

Since the collinear approximation keeps the two spin channels' density matrices independent, $\rho_\downarrow$ and $\sigma_{\downarrow\downarrow}$ do not depend on $D^\uparrow$, and only three chain-rule terms survive.

2 Step 1: the density gradient as a linear map of D

By the quotient rule, $\rho' = q'/(4\pi r^2) - 2\rho/r$. Because D is symmetric, relabeling the dummy indices $i \leftrightarrow j$ in half of the product-rule expansion collapses the sum:

$$q'(r) = \sum_{ij} D_{ij} (\varphi'_i \varphi_j + \varphi_i \varphi'_j) = 2 \sum_{ij} D_{ij} \varphi'_i \varphi_j. \quad (4)$$

Differentiating ρ' with respect to a single (unsymmetrized) entry D_{ij} – the mass-matrix building blocks introduced in Step 3 restore the correct symmetric matrix, so no care is needed here about double-counting off-diagonal entries – gives, using (4) and the $-2\rho/r$ term (which itself has $\partial\rho/\partial D_{ij} = \varphi_i\varphi_j/4\pi r^2$):

$$\frac{\partial\rho'(r)}{\partial D_{ij}} = \frac{1}{4\pi r^2} \left[\varphi'_i\varphi_j + \varphi_i\varphi'_j - \frac{2}{r}\varphi_i\varphi_j \right]. \quad (5)$$

The $-2/r$ term is a purely radial-reduction artifact: it comes from $q = 4\pi r^2\rho$, not from any physical content of the GGA functional, and would be absent in a Cartesian weak-form derivation.

3 Step 2: chain rule through $\rho_\uparrow$, $\sigma_{\uparrow\uparrow}$, $\sigma_{\uparrow\downarrow}$

$$\frac{\partial E_{xc}}{\partial D_{ij}^\uparrow} = \int_0^\infty 4\pi r^2 \left[v_{\rho\uparrow} \frac{\partial\rho_\uparrow}{\partial D_{ij}^\uparrow} + v_{\sigma\uparrow\uparrow} \frac{\partial\sigma_{\uparrow\uparrow}}{\partial D_{ij}^\uparrow} + v_{\sigma\uparrow\downarrow} \frac{\partial\sigma_{\uparrow\downarrow}}{\partial D_{ij}^\uparrow} \right] dr. \quad (6)$$

Using $\sigma_{\uparrow\uparrow} = \rho_\uparrow^2$ and $\sigma_{\uparrow\downarrow} = \rho'_\uparrow\rho'_\downarrow$ (product rule):

$$\frac{\partial\sigma_{\uparrow\uparrow}}{\partial D_{ij}^\uparrow} = 2\rho'_\uparrow \frac{\partial\rho'_\uparrow}{\partial D_{ij}^\uparrow}, \quad \frac{\partial\sigma_{\uparrow\downarrow}}{\partial D_{ij}^\uparrow} = \rho'_\downarrow \frac{\partial\rho'_\uparrow}{\partial D_{ij}^\uparrow}. \quad (7)$$

Both multiply the *same* factor $\partial\rho'_\uparrow/\partial D_{ij}^\uparrow$ from (5), so they combine into a single coefficient:

$$K_\uparrow(r) := 2v_{\sigma\uparrow\uparrow}(r) \rho'_\uparrow(r) + v_{\sigma\uparrow\downarrow}(r) \rho'_\downarrow(r). \quad (8)$$

Substituting (5)–(8) and $\partial\rho_\uparrow/\partial D_{ij}^\uparrow = \varphi_i\varphi_j/4\pi r^2$, the factors of $4\pi r^2$ cancel exactly:

$$\frac{\partial E_{xc}}{\partial D_{ij}^\uparrow} = \int_0^\infty \left\{ v_{\rho\uparrow} \varphi_i\varphi_j + K_\uparrow \left[\varphi'_i\varphi_j + \varphi_i\varphi'_j - \frac{2}{r}\varphi_i\varphi_j \right] \right\} dr \quad (9)$$

$$= \int_0^\infty \left[v_{\rho\uparrow} - \frac{2}{r}K_\uparrow \right] \varphi_i\varphi_j dr + \int_0^\infty K_\uparrow \varphi'_i\varphi_j dr + \int_0^\infty K_\uparrow \varphi_i\varphi'_j dr. \quad (10)$$

4 Step 3: identifying the matrix building blocks

Define the weighted mass matrix and the (asymmetric) mixed mass matrix already used elsewhere in the codebase for the LDA and Hartree assembly:

$$M[w]_{ij} := \int_0^\infty w(r) \varphi_i(r)\varphi_j(r) dr \quad (\text{symmetric}), \quad Mmix[w]_{ij} := \int_0^\infty w(r) \varphi'_i(r)\varphi_j(r) dr \quad (\text{not symmetric}). \quad (11)$$

$M[w]$ is symmetric because $\varphi_i\varphi_j$ does not distinguish the order of i, j ; $Mmix[w]$ is not, since $\varphi'_i\varphi_j \neq \varphi'_j\varphi_i$ in general. The third integral in (10) is exactly the transpose of the second under this definition:

$$\int_0^\infty K_\uparrow \varphi_i\varphi'_j dr = \int_0^\infty K_\uparrow \varphi'_j\varphi_i dr = Mmix[K_\uparrow]_{ji} = (Mmix[K_\uparrow]^T)_{ij}. \quad (12)$$

Hence, in matrix form:

$$\boxed{V_{xc\uparrow} = M \left[v_{\rho\uparrow} - \frac{2}{r}K_\uparrow \right] + Mmix[K_\uparrow] + Mmix[K_\uparrow]^T}, \quad K_\uparrow = 2v_{\sigma\uparrow\uparrow}\rho'_\uparrow + v_{\sigma\uparrow\downarrow}\rho'_\downarrow \quad (13)$$

and $V_{xc\downarrow}$, $K_\downarrow$ follow by exchanging $\uparrow \leftrightarrow \downarrow$ throughout. The $Mmix[K_\uparrow] + Mmix[K_\uparrow]^T$ combination is what makes $V_{xc\uparrow}$ symmetric despite each piece individually not being so – physically required of any Hamiltonian matrix, and structurally the same defensive $(V + V^T)/2$ pattern already used in `assemble_exc!`.

Proposition 4.1 (Consistency check: ∇ -independent limit). *Setting $v_{\sigma\alpha\beta} \equiv 0$ (an LDA functional has no gradient dependence) collapses $K_{\uparrow} \rightarrow 0$ and (13) reduces to $V_{xc\uparrow} = M[v_{\rho\uparrow}]$, the ordinary LDA weighted mass matrix – as it must.*

5 Step 4: the unpolarized case, $n_{spin} = 1$

An unpolarized GGA functional depends on a single gradient invariant $\sigma = \rho'^2$ rather than the three spin-resolved invariants above – there is no second channel to mix with, so no cross term arises. Repeating Steps 1–3 with $\sigma_{\uparrow\uparrow} \rightarrow \sigma$, $\sigma_{\uparrow\downarrow} \rightarrow 0$ gives simply

$$K = 2v_{\sigma} \rho', \quad V_{xc} = M\left[v_{\rho} - \frac{2}{r}K\right] + Mix[K] + Mix[K]^T = M\left[v_{\rho} - \frac{4}{r}v_{\sigma}\rho'\right] + Mix[2v_{\sigma}\rho'] + Mix[2v_{\sigma}\rho']^T, \quad (14)$$

matching the single-channel formula originally derived in `dev/gga/PLAN.md`.